

Variational Geometry of Optimal Pandrosion Charts

Curvature Energies, Euler Directions, and Finite-Dimensional Chart Descent

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Abstract

Anchored divided-difference iteration has a local multiplier controlled by the logarithmic curvature of the transformed equation $F_\phi = G \circ \phi$. Earlier chart optimization cancels the pointwise quadratic defect $F_\phi''(r) = 0$, and higher-order inverse jets flatten $G \circ \phi$ to arbitrary finite order. This paper proposes a variational continuation: instead of canceling only a finite jet at one root, choose charts by minimizing a curvature energy

$$\mathcal{E}(\phi) = \int_D \left| \frac{F_\phi''(s)}{F_\phi'(s)} \right|^2 w(s) ds.$$

The integrand is the square of the charted logarithmic curvature

$$\kappa_\phi = (\log F_\phi')' = \frac{G'''(\phi)}{G'(\phi)} \phi' + \frac{\phi''}{\phi'}.$$

Local curvature cancellation is the pointwise condition $\kappa_\phi(r) = 0$, while the exact inverse chart makes $\kappa_\phi \equiv 0$. In constrained chart families, such as polynomial jets, variational minimization gives a finite-dimensional normal equation for the best available flattening. Figures for $G(z) = e^z - 2$ show the curvature-energy landscape in a quadratic-cubic family, the gradient-descent path toward inverse-jet coefficients, and the decay of energy along higher inverse-jet truncations. The central message is that optimal chart selection can be formulated as a variational geometry of analytic coordinates.

One-sentence summary. The scalar chart-selection law becomes a variational principle: minimize the L^2 energy of the logarithmic curvature $\kappa_\phi = F_\phi''/F_\phi'$.

1 Introduction

The anchored divided-difference multiplier near a simple root is

$$P'_{\phi,a}(r) = \frac{F_\phi''(r)}{2F_\phi'(r)}(a - r) + \mathcal{O}((a - r)^2).$$

Thus the ratio

$$\kappa_\phi(r) = \frac{F_\phi''(r)}{F_\phi'(r)}$$

is the local curvature that controls the residual. The optimal-chart paper cancels this curvature at a point. The higher-order jet paper cancels more Taylor coefficients at the same point.

The present paper asks a different question: can chart selection be made variational? Instead of imposing one pointwise constraint, we minimize an energy measuring how nonlinear the transformed equation is over a neighborhood.

2 The logarithmic curvature of a chart

Let G be analytic near a simple zero ρ , and let ϕ be a local chart. Set

$$F_\phi(s) = G(\phi(s)).$$

Assume $F'_\phi(s) \neq 0$ on a small domain D .

Definition 2.1 (Chart curvature). The logarithmic curvature of the charted equation is

$$\kappa_\phi(s) = \frac{F''_\phi(s)}{F'_\phi(s)} = (\log F'_\phi(s))'. \quad (1)$$

By the chain rule,

$$\kappa_\phi(s) = \frac{G''(\phi(s))}{G'(\phi(s))} \phi'(s) + \frac{\phi''(s)}{\phi'(s)}. \quad (2)$$

This is the same expression that appeared in the local chart multiplier law.

Proposition 2.2 (Pointwise curvature cancellation). *Let r be the root coordinate. A root-normalized chart satisfies*

$$P'_{\phi,a}(r) = \frac{1}{2} \kappa_\phi(r)(a - r) + \mathcal{O}((a - r)^2).$$

Consequently, first-order local optimality is equivalent to

$$\kappa_\phi(r) = 0.$$

Proof. This is the local multiplier expansion applied to F_ϕ . □

3 Curvature energy

Definition 3.1 (Variational chart energy). Let $D \subset \mathbb{C}$ be a small chart domain and let $w \geq 0$ be a weight. Define

$$\mathcal{E}_D(\phi) = \int_D |\kappa_\phi(s)|^2 w(s) dA(s). \quad (3)$$

On a real interval I , the corresponding one-dimensional energy is

$$\mathcal{E}_I(\phi) = \int_I |\kappa_\phi(s)|^2 w(s) ds.$$

This energy vanishes exactly when F_ϕ is affine on the domain. If root normalization is imposed, the affine model is $F_\phi(s) = s - r$.

Theorem 3.2 (Exact inverse charts are zero-energy minimizers). *Let G be locally invertible near a simple zero ρ . If $\phi_*(s) = G^{-1}(s)$ in a neighborhood of 0, then*

$$G(\phi_*(s)) = s, \quad \kappa_{\phi_*} \equiv 0, \quad \mathcal{E}(\phi_*) = 0.$$

Conversely, if $\mathcal{E}(\phi) = 0$ on a connected domain and F'_ϕ is nonzero, then F_ϕ is affine. Under root normalization, $F_\phi(s) = s$.

Proof. The inverse chart statement is immediate. Conversely, $\mathcal{E}(\phi) = 0$ implies $\kappa_\phi = 0$ almost everywhere, hence everywhere by analyticity. Therefore $(\log F'_\phi)' = 0$, so F'_ϕ is constant and F_ϕ is affine. Root normalization fixes the affine map to be the identity. □

4 Finite-dimensional chart families

In practice one does not optimize over all analytic charts. One chooses a finite-dimensional family

$$\phi_\theta(s) = \rho + \frac{s}{G'(\rho)} + \sum_{j=2}^m \theta_j s^j.$$

Then

$$\mathcal{E}(\theta) = \int_I |\kappa_{\phi_\theta}(s)|^2 w(s) ds.$$

Proposition 4.1 (Normal equations in a chart family). *Assume $\theta \mapsto \phi_\theta$ is differentiable and I is real. A stationary point of $\mathcal{E}$ satisfies*

$$\operatorname{Re} \int_I \overline{\kappa_\theta(s)} \frac{\partial \kappa_\theta(s)}{\partial \theta_j} w(s) ds = 0 \quad (j = 1, \dots, N). \quad (4)$$

Proof. Differentiate $\mathcal{E}(\theta) = \int |\kappa_\theta|^2 w$ under the integral sign:

$$\frac{\partial \mathcal{E}}{\partial \theta_j} = 2 \operatorname{Re} \int \overline{\kappa_\theta} \frac{\partial \kappa_\theta}{\partial \theta_j} w.$$

Setting the derivative to zero gives (4). □

Remark 4.2. The normal equations are the variational analogue of the jet cancellation equations. Jet cancellation imposes vanishing Taylor coefficients at one point. Energy minimization distributes the flattening over a domain.

5 Example: $G(z) = e^z - 2$

Let $\rho = \log 2$ and consider the real chart family

$$\phi_{\eta, \zeta}(s) = \rho + \frac{s}{2} + \eta s^2 + \zeta s^3.$$

For $G(z) = e^z - 2$, one has $G''/G' = 1$, so

$$\kappa_{\eta, \zeta}(s) = \phi'_{\eta, \zeta}(s) + \frac{\phi''_{\eta, \zeta}(s)}{\phi'_{\eta, \zeta}(s)}.$$

The exact inverse is

$$\log(2 + s) = \rho + \frac{s}{2} - \frac{s^2}{8} + \frac{s^3}{24} - \dots$$

Thus the quadratic-cubic inverse jet has

$$\eta_* = -\frac{1}{8}, \quad \zeta_* = \frac{1}{24}.$$

We use the weighted interval energy

$$\mathcal{E}(\eta, \zeta) = \frac{\int_{-0.65}^{0.65} |\kappa_{\eta, \zeta}(s)|^2 e^{-(s/0.55)^2} ds}{\int_{-0.65}^{0.65} e^{-(s/0.55)^2} ds}.$$

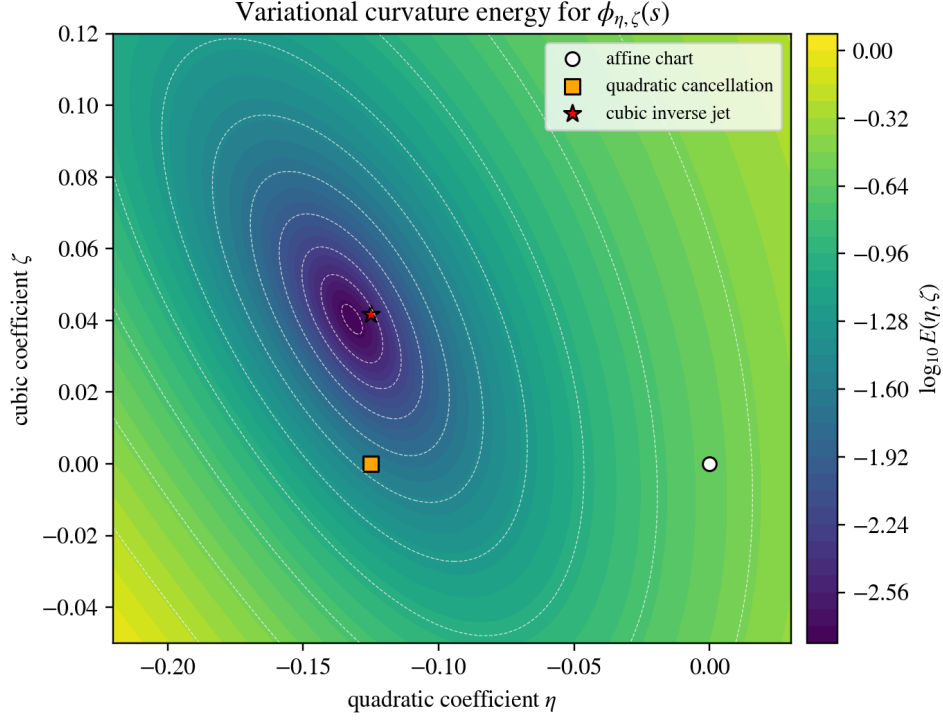


Figure 1: Curvature-energy landscape for the quadratic-cubic chart family. The affine chart, quadratic cancellation, and cubic inverse jet are marked. The inverse-jet coefficients lie near the low-energy valley.

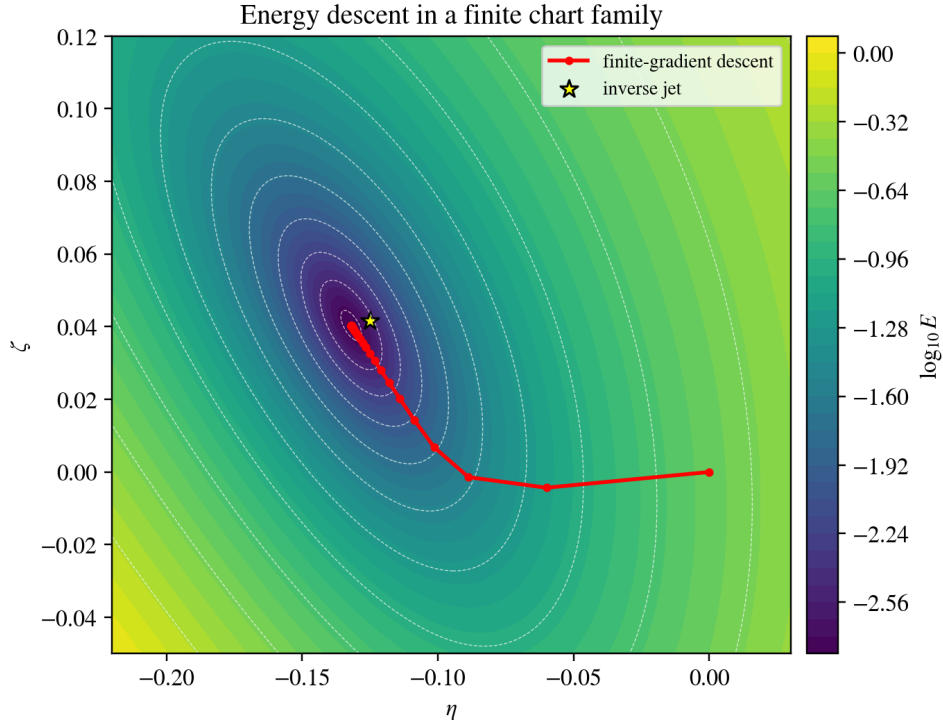


Figure 2: Finite-gradient descent from the affine chart in the same energy landscape. The descent moves toward the inverse-jet valley.

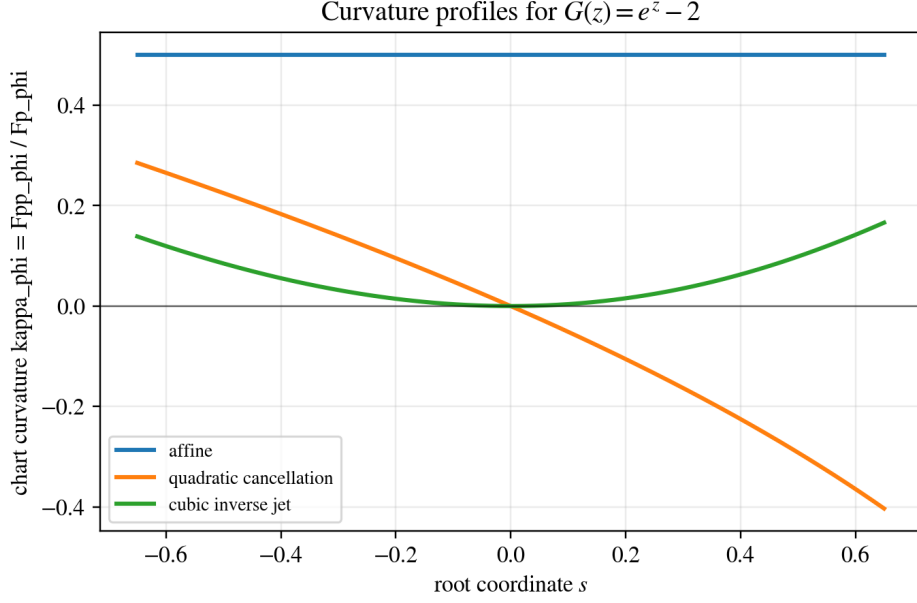


Figure 3: Curvature profiles $\kappa_\phi = F''_\phi / F'_\phi$ on the real root coordinate. The quadratic and cubic inverse-jet charts suppress curvature near the root compared with the affine chart.

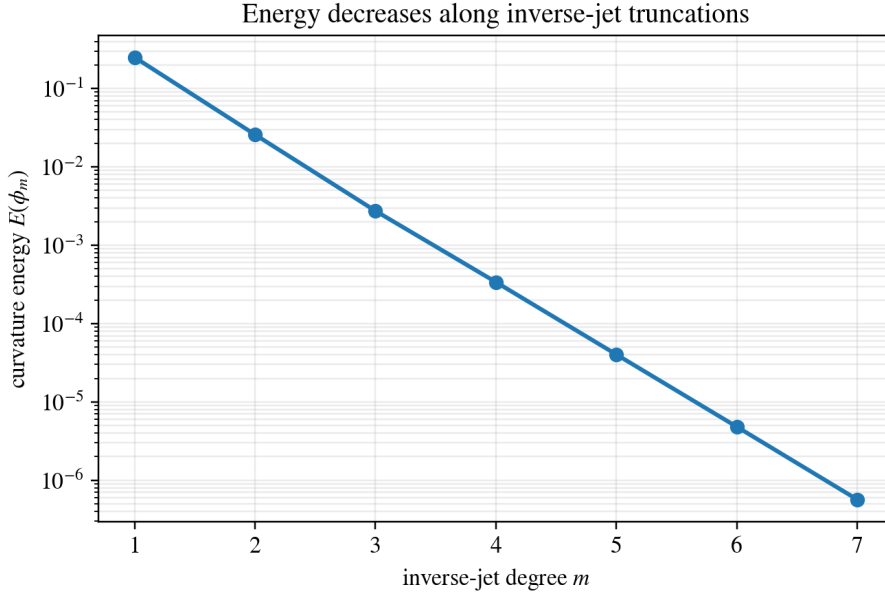


Figure 4: Curvature energy along inverse-jet truncations. Increasing the inverse-jet degree reduces the energy over the chosen interval until numerical and truncation effects dominate.

6 Relation with higher-order jet cancellation

Higher-order jet cancellation is a local normal-form statement:

$$G(\phi_m(s)) = s + \mathcal{O}(s^{m+1}).$$

Variational chart geometry replaces this pointwise order condition by an integral criterion. The two viewpoints agree in the limit of a shrinking weight. If w_ϵ concentrates near the root,

then minimizing $\mathcal{E}_{w_\epsilon}$ forces successive Taylor coefficients of κ_ϕ to vanish, recovering inverse-jet cancellation.

Proposition 6.1 (Concentration recovers local cancellation). *Let $w_\epsilon(s) = \epsilon^{-1}w(s/\epsilon)$ on a real interval and assume a polynomial chart family contains the inverse jet through degree m . As $\epsilon \rightarrow 0$, any minimizer of $\mathcal{E}_{w_\epsilon}$ whose coefficients remain bounded satisfies the inverse-jet cancellation equations through degree m , provided the minimizer is isolated.*

Proof. Expanding $\kappa_\phi(s)$ in a Taylor series, the concentrated energy is asymptotically a weighted sum of squared Taylor coefficients with powers of ϵ . The leading nonzero coefficient dominates. Minimization therefore cancels the lowest available coefficient first, then the next, and so on. This is exactly the inverse-jet hierarchy. $\square$

7 Outlook

The variational viewpoint suggests several extensions:

- (i) rational chart families, where Padé-type inverse approximants can reduce energy on larger domains;
- (ii) complex-area energies over disks, annuli, or basin neighborhoods;
- (iii) multidimensional tensor energies such as $\int \|D^2(G \circ \phi)\|^2$;
- (iv) adaptive engines that update chart parameters by residual-gated energy descent.

8 Conclusion

The local chart-selection law can be globalized into a variational principle. The quantity

$$\kappa_\phi = \frac{F''_\phi}{F'_\phi}$$

is both the local multiplier coefficient and the curvature of the transformed equation. Minimizing its energy gives a principled way to choose charts when exact inverse charts are unavailable. This reframes chart selection as a variational geometry of analytic coordinates.

References

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